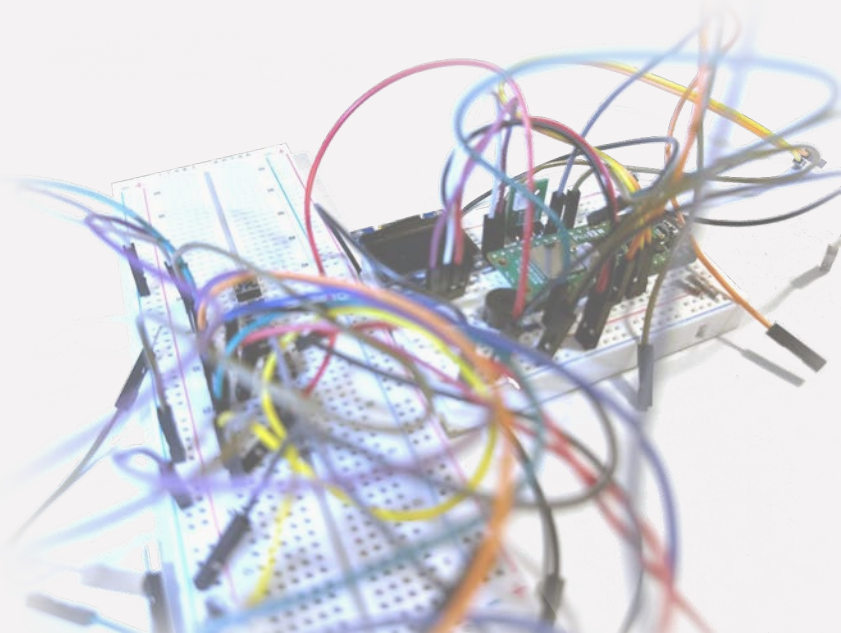
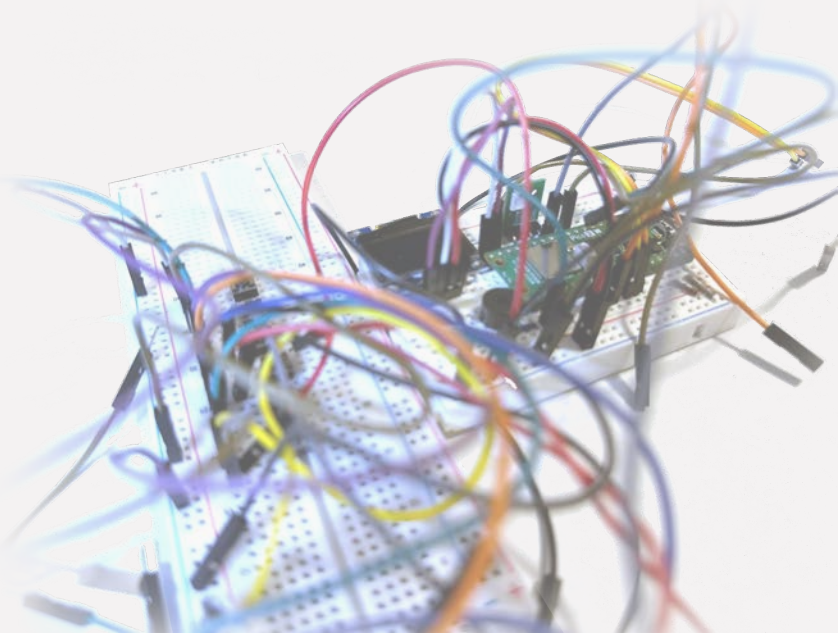
Inventor Pi

Introduction to Electronics: Class Experiments Kit

Utshaho Gupta



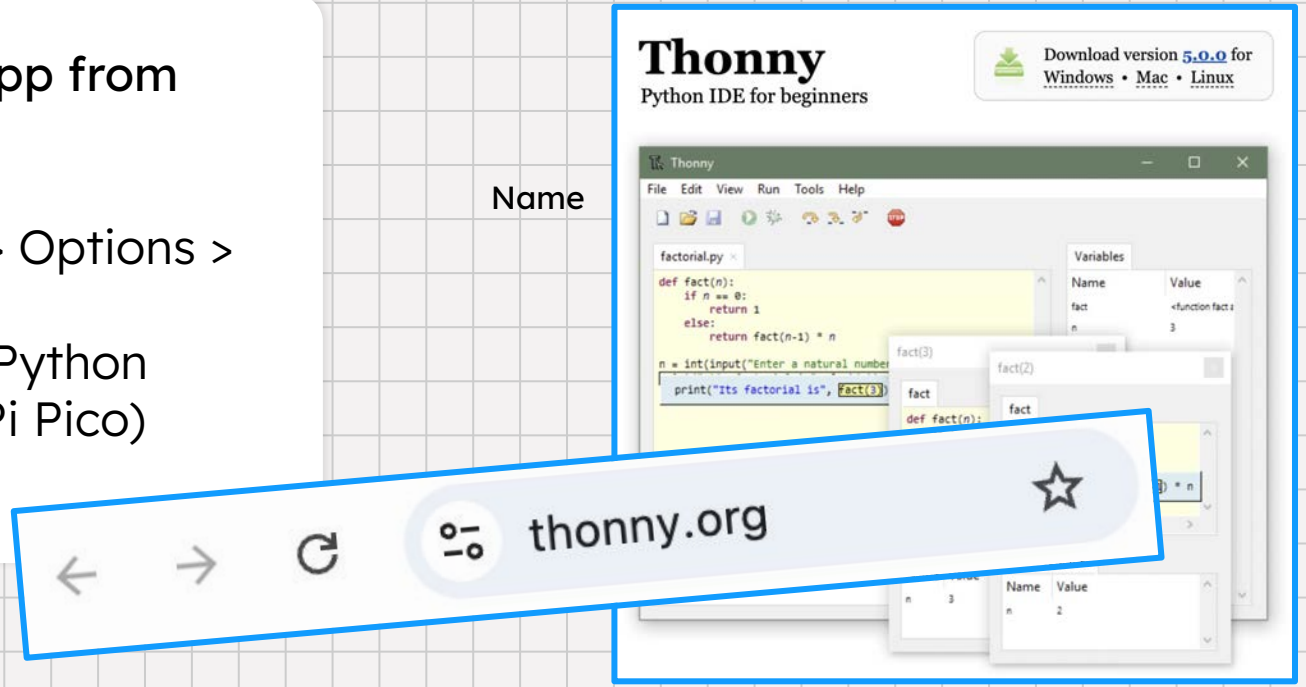
With the right guidance, everyone can learn electronics, AI, and computer vision through projects that connect to real-world impact.



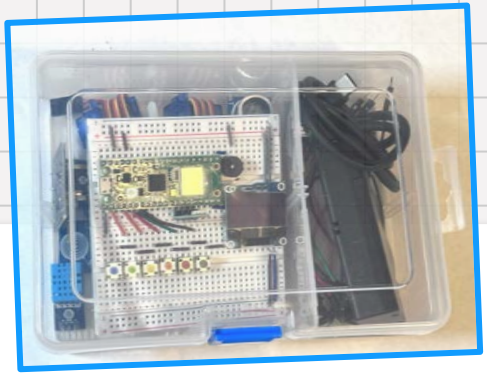
Install Thonny app

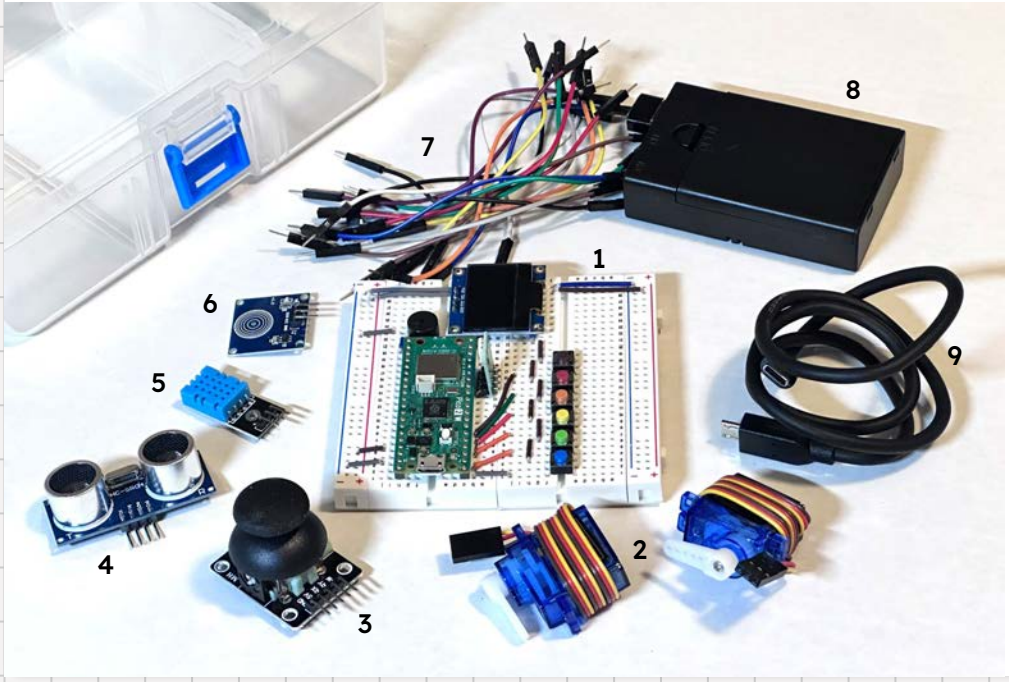
- Install Thonny app from **thonny.org**
- Open Thonny
  - Go to Tools > Options > Interpreter
  - Select MicroPython (Raspberry Pi Pico)
  - Click OK

Name



Kit Contents

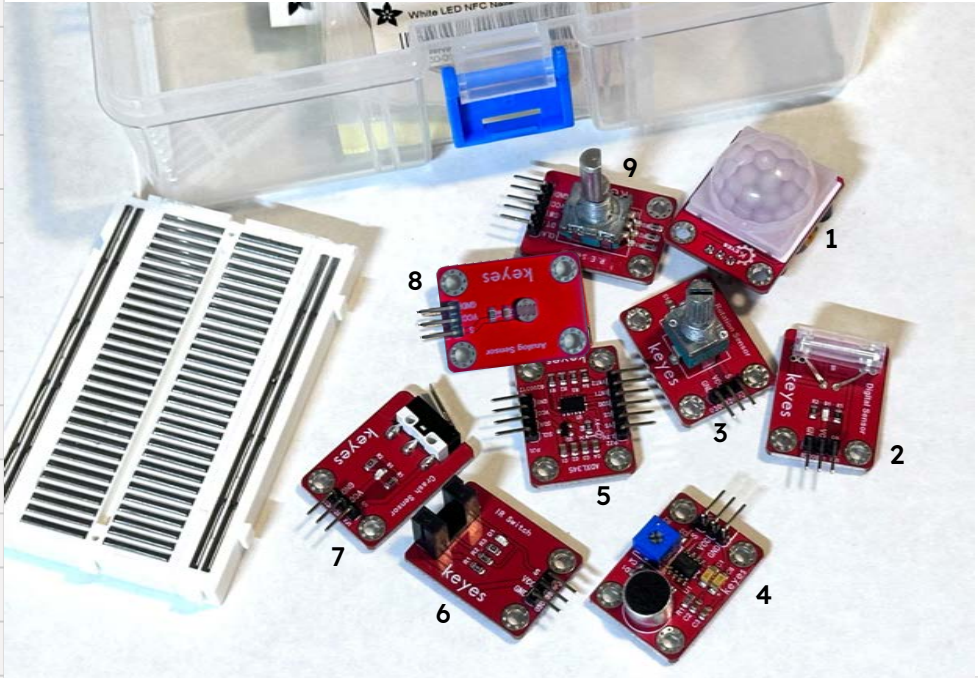




1. Breadboard with OLED, LED, buzzer, Raspberry Pi Pico 2 W, 6 buttons
2. 2 Micro servo motors
3. Joystick module
4. Distance sensor
5. Humidity + temperature sensor
6. Touch sensor
7. 20 jumper wires
8. Battery pack
9. Micro USB cable

Additional Sensors

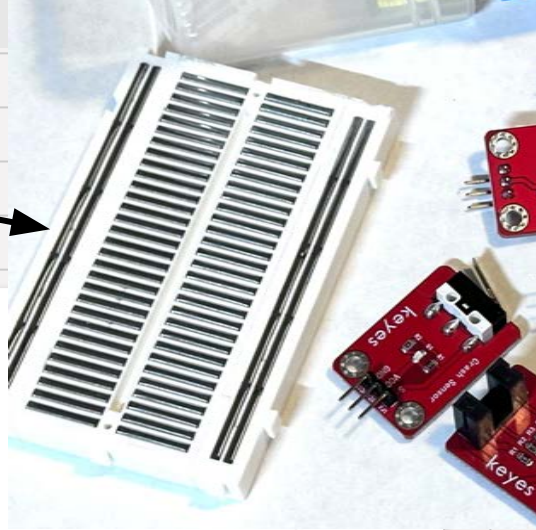




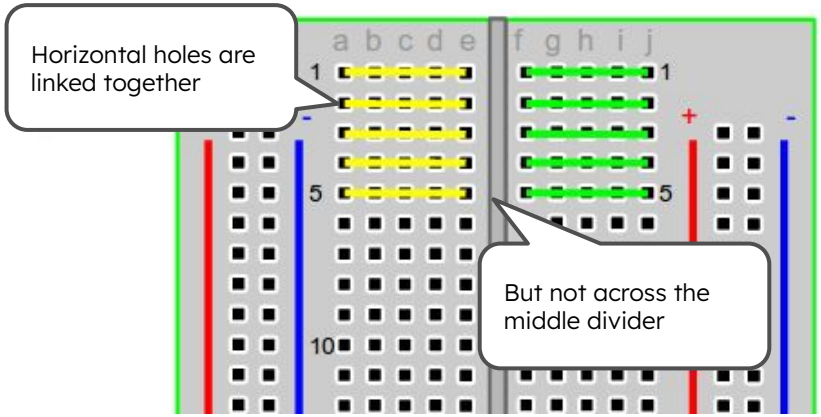
Additional Sensors for Sharing

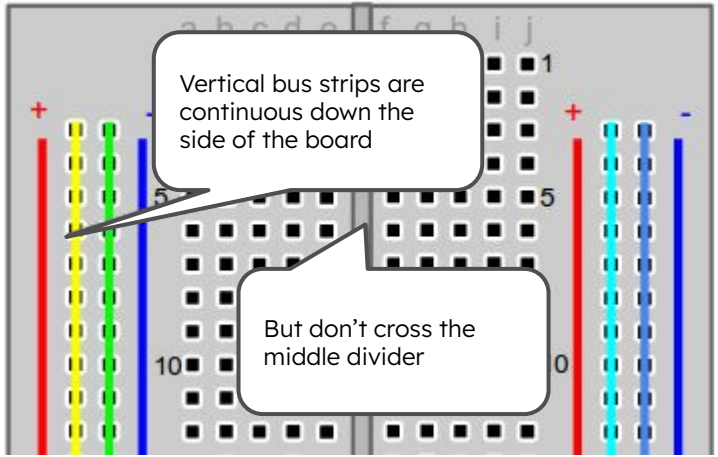
1. Motion sensor
2. Knock sensor
3. Potentiometer
4. Microphone sensor
5. Accelerometer
6. Photo interrupter
7. Crash sensor
8. Photoresistor
9. Rotation Encoder

Breadboard Wiring



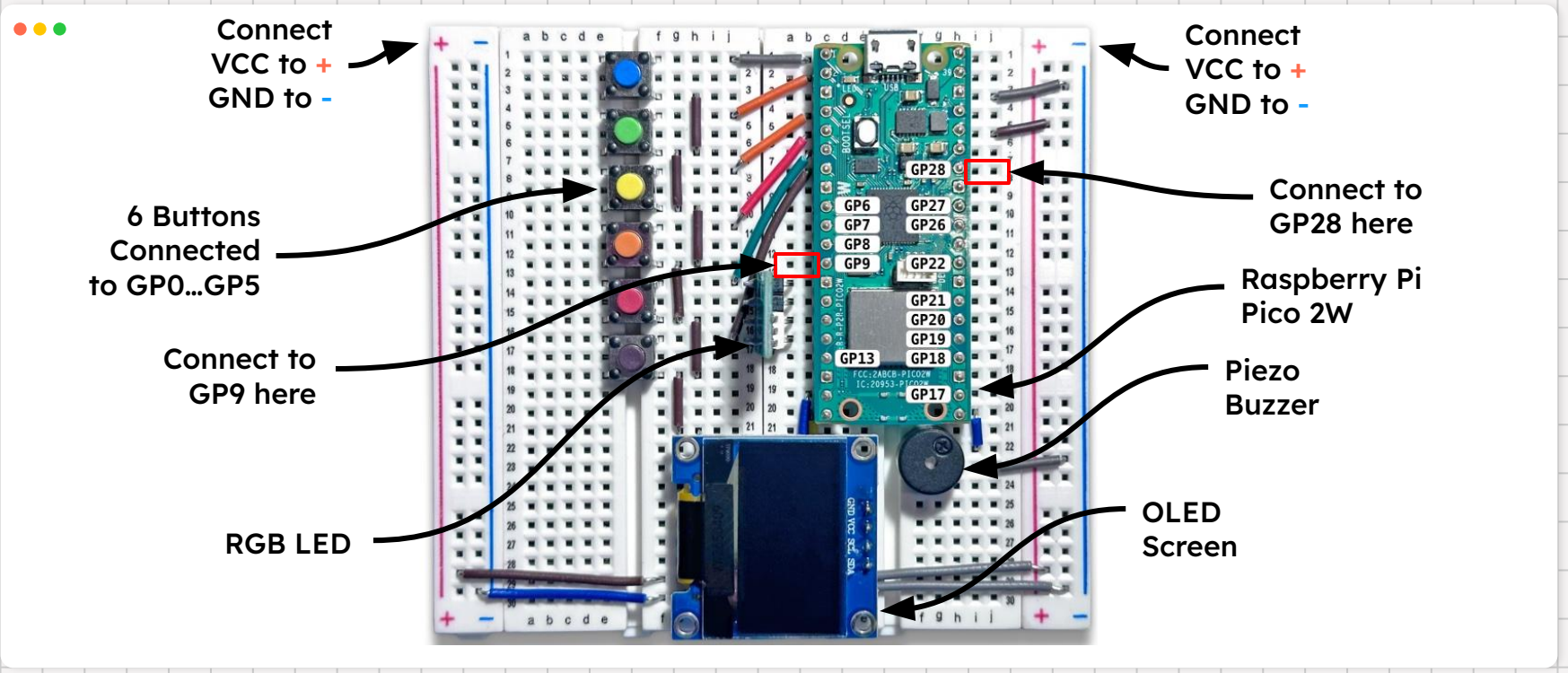
You can see the connections behind a breadboard



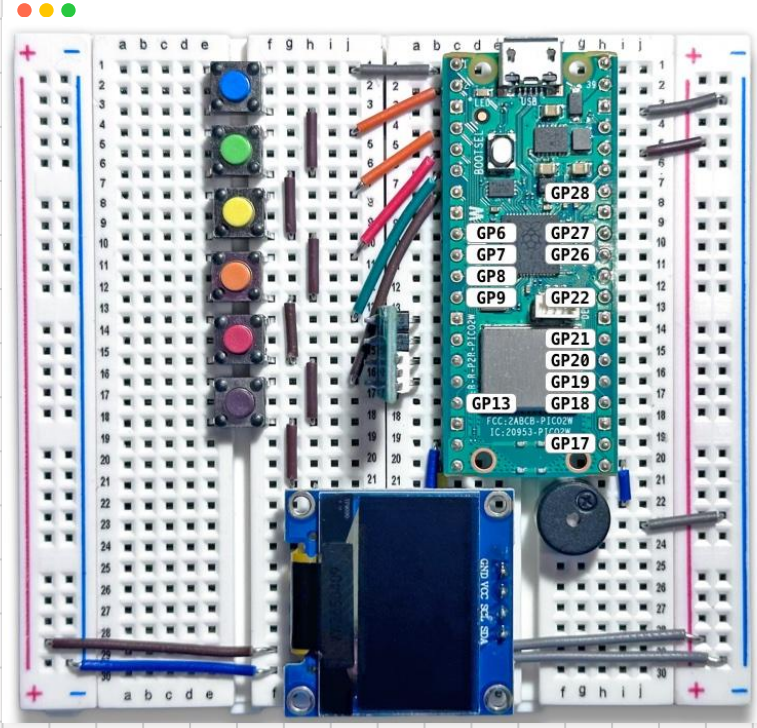


Breadboard is a solderless way to connect electronic components for prototyping

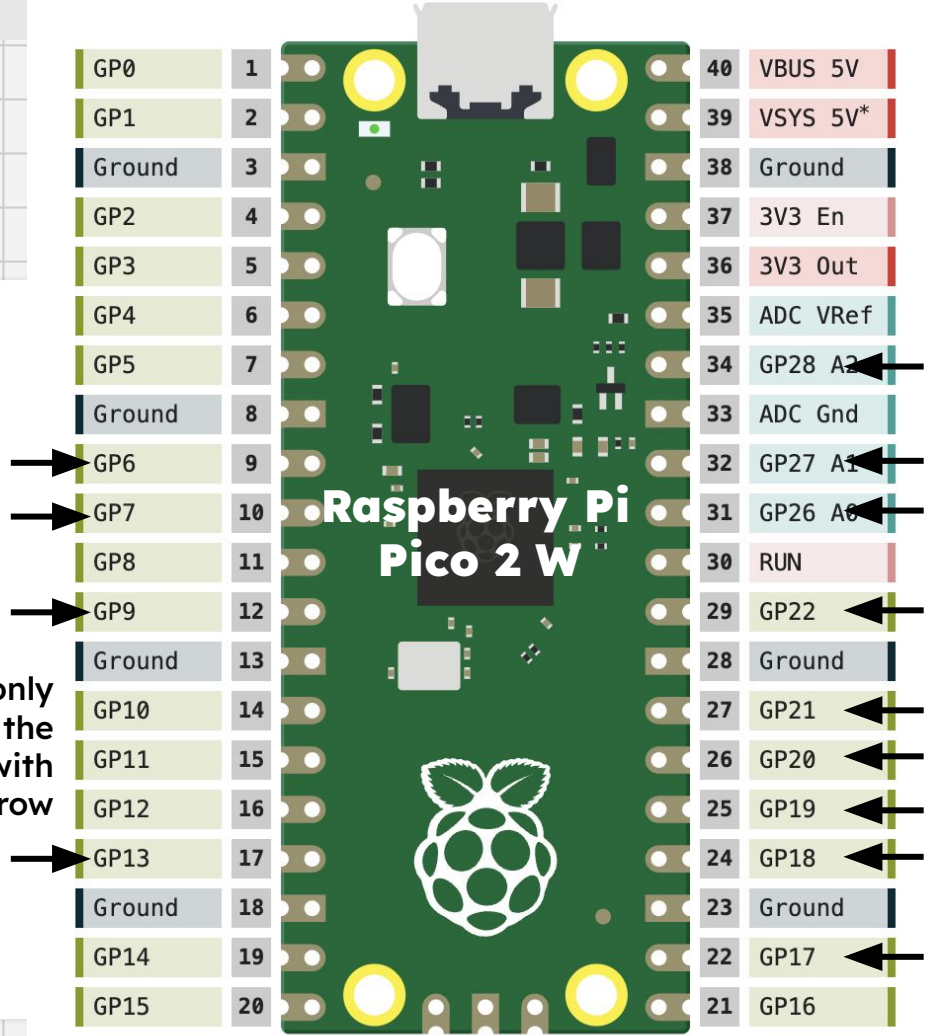
Breadboard Wiring



GPIO Pinout



We'll only use the pins with an arrow



Important Do’s and Don’ts

Do’s

1. Disconnect Pico before changing connectors
2. Verify all connections, especially VCC and GND
3. Establish team roles like coding, wiring, verifying circuit, capturing notes and recording
4. Read experiment description and code

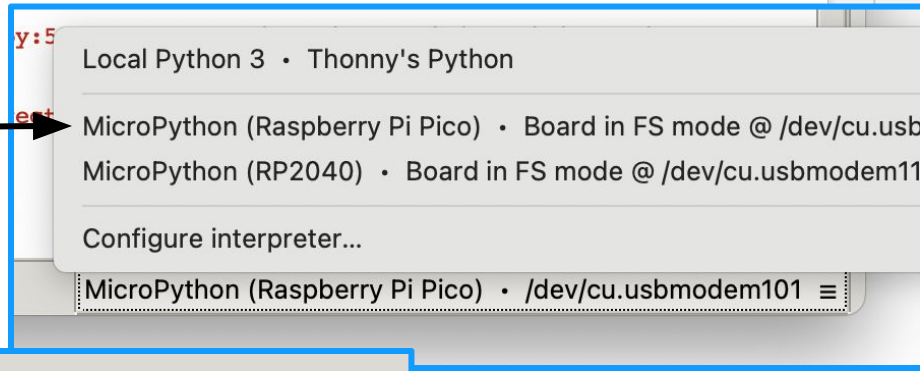
Don’ts

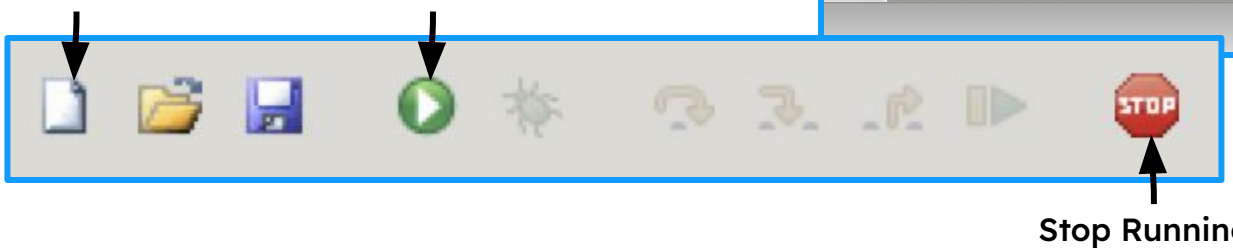
1. Don’t power on before verifying all connections. Incorrect VCC and GND can destroy sensitive electronics
2. Don’t rely on numbers printed on the breadboard. Count point distance from other connectors on the board to verify
3. Don’t leave before putting everything back in the kit box

Connect Pico



1. Launch Thonny on computer
2. Connect Pico to computer via USB
3. Click ≡ icon on bottom right
4. Select Raspberry Pi Pico





Code

Reading example code

1. First read description
2. Connect any components
3. Read the main loop
4. Read other comments
5. Read rest of the code

crash_sensor.py

```
print("\nCrash Sensor")

# DESCRIPTION

# A crash sensor uses a limit switch. This can be used with a hard stop to detect
# when a mechanism, such as a robotic arm, reaches the limit of its motion and
# trigger a limit switch. It is also used in oven and refrigerator doors to turn
# on the internal light when the door is opened.

# CONNECTIONS REQUIRED

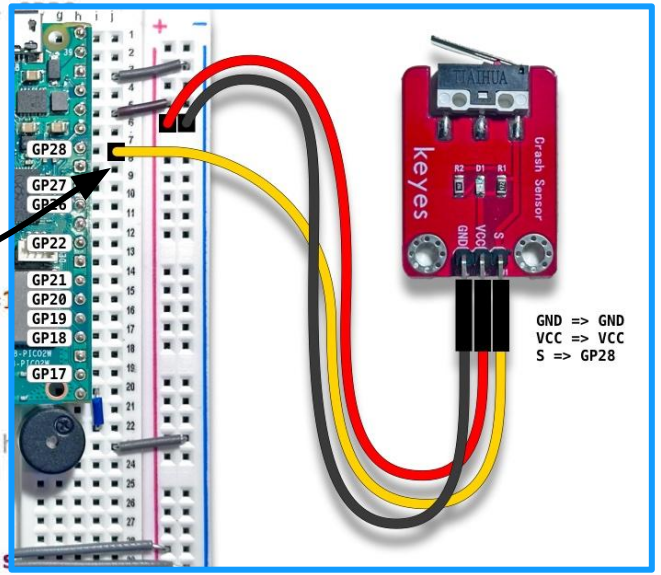
# Connect Crash Sensor (KY-021) module
#   Pins: GND => GND, VCC => VCC, S =>

from machine import Pin
from oled import OLED
from buzzer import Buzzer
from rgbled import RGBLED

sensor = Pin(28, Pin.IN, Pin.PULL_UP)
oled = OLED(scl_pin=15, sda_pin=14)
buzzer = Buzzer(pwm_pin=16)
rgbled = RGBLED(red_pin=12, green_pin=1

try:
    while True:

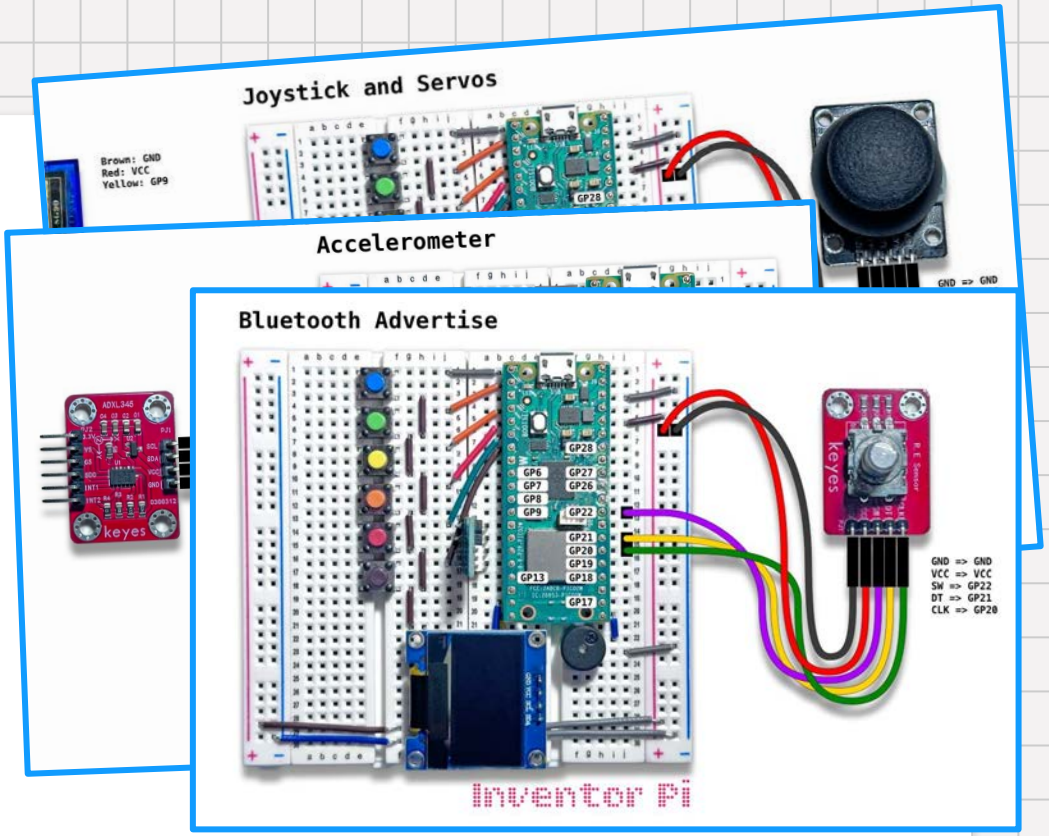
        # Sensor value is 1 => No crash
        # Loop while no crash detected
        oled.reset()
        rgbled.set_color(g=255)
        while sensor.value() == 1: pass
```



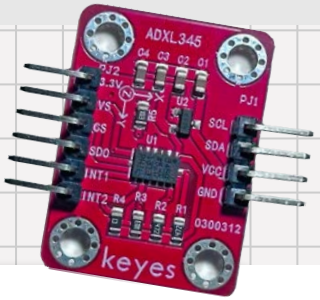
Experiments

1. Write first script: ***blink.py***
2. Connect first sensor Humidity Temp Sensor. Write ***humidity_temp_sensor.py***
3. Now choose another experiment at https://github.com/goatoush/inventor-pi

Each team picks 1 experiment at a time. After a successful demo to the class, you can pick another experiment.



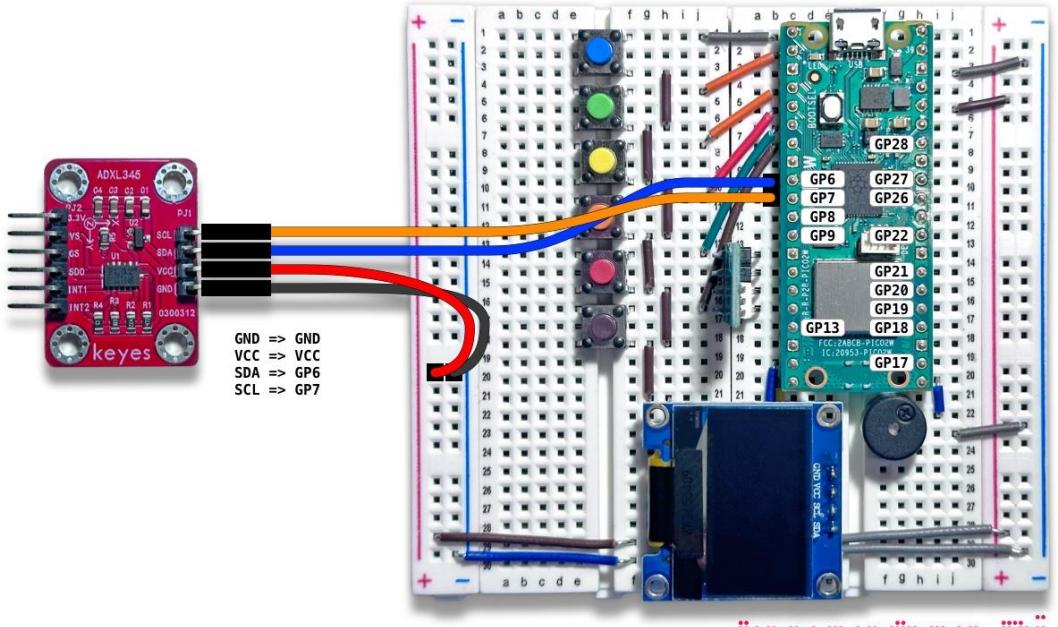
Example: Accelerometer



An accelerometer measures acceleration around the x, y and z axis. In robotics, it is essential in determining the robot position and movement.

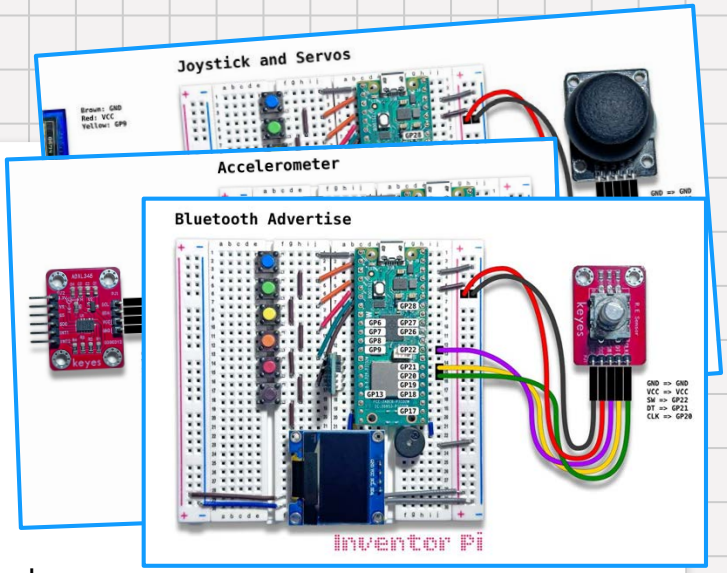
We will draw a 3D shape, and control its rotation using the accelerometer motion.

Accelerometer

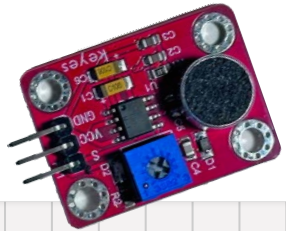


Experiment List

1. Blink
2. Humidity Temp Sensor
3. Melody
4. Touch Game
5. Distance Sensor
6. Piano
7. Joystick and Servos
8. Microphone Sensor and Melody
9. Accelerometer
10. Crash Sensor
11. Knock Sensor
12. Motion Sensor
13. Photo Interrupter
14. Photoresistor
15. Potentiometer and Servo
16. Bluetooth Advertise and Bluetooth Scan
17. Mix and match inputs and outputs









Continue Learning



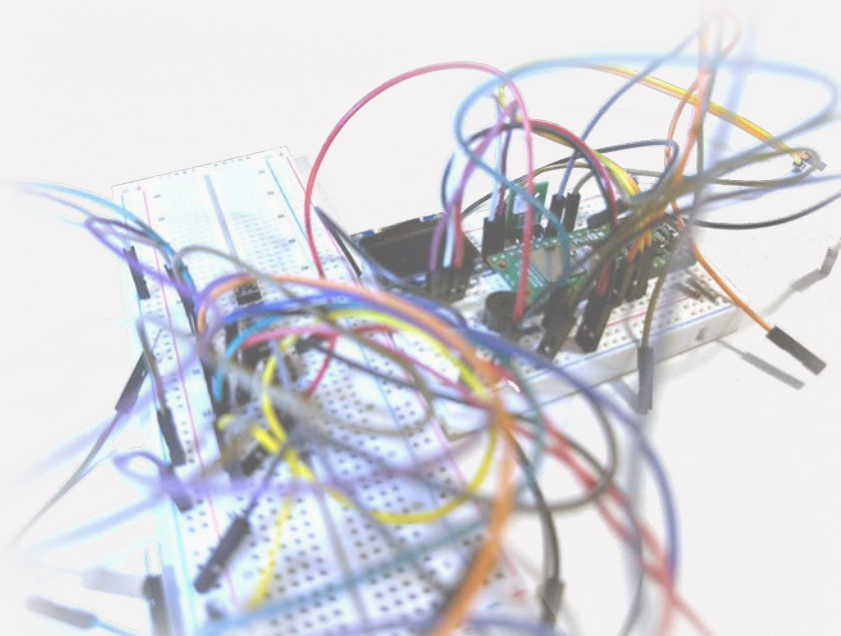
We focused on learning the building blocks. You saw how easy it is to start building.



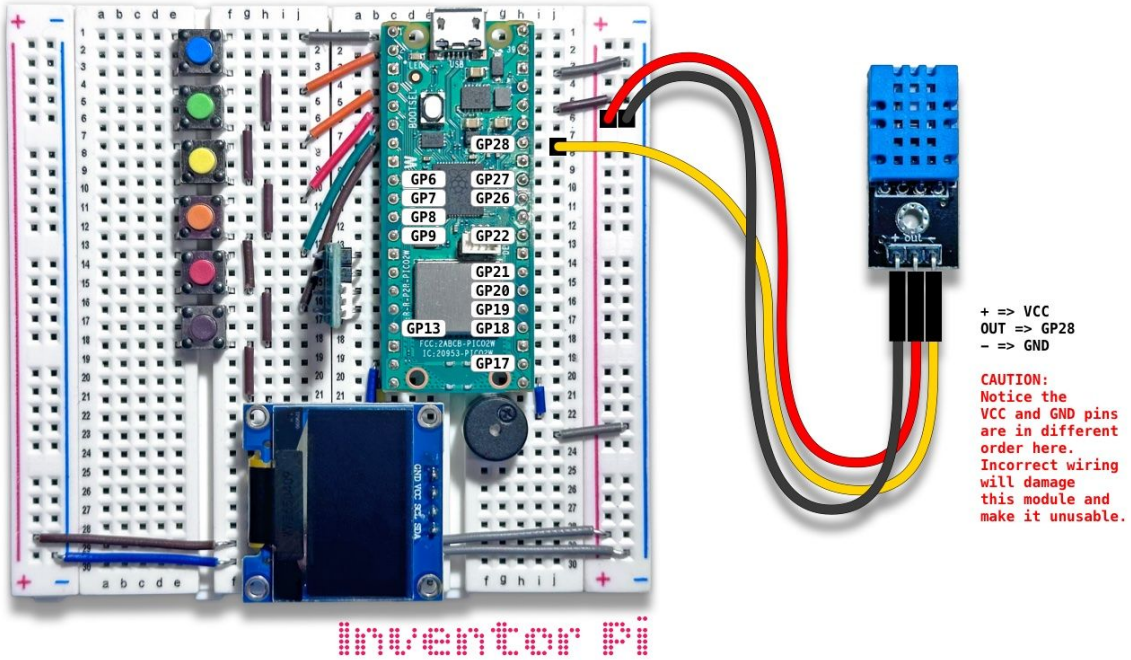
Now it's your turn to prototype your ideas. Identify real world problems, and continue learning new skills, experimenting and **inventing**.

Email with your progress and any questions at ugupta@aischool.org

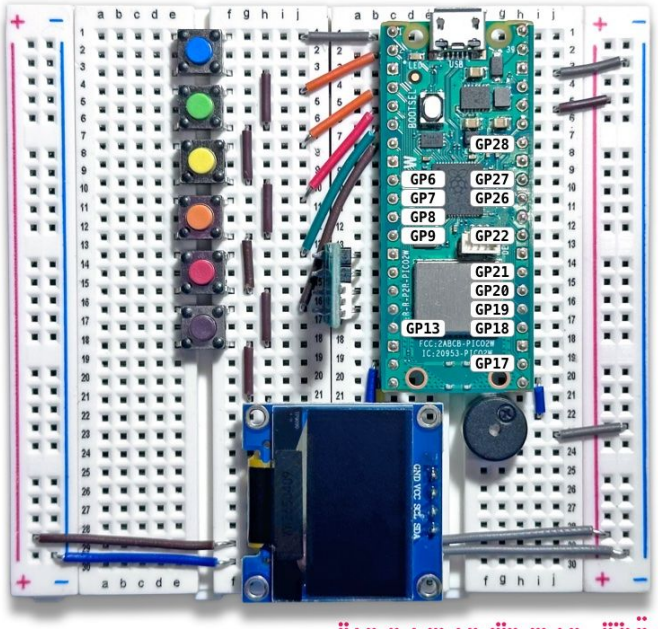
Experiment Circuit Diagrams for Reference



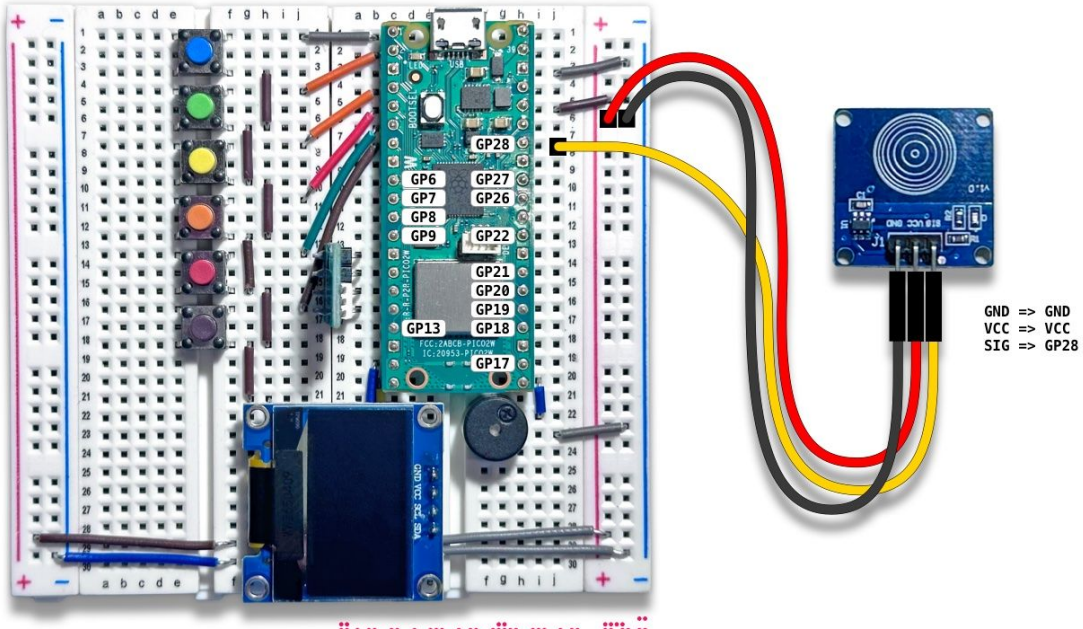
Humidity Temp Sensor



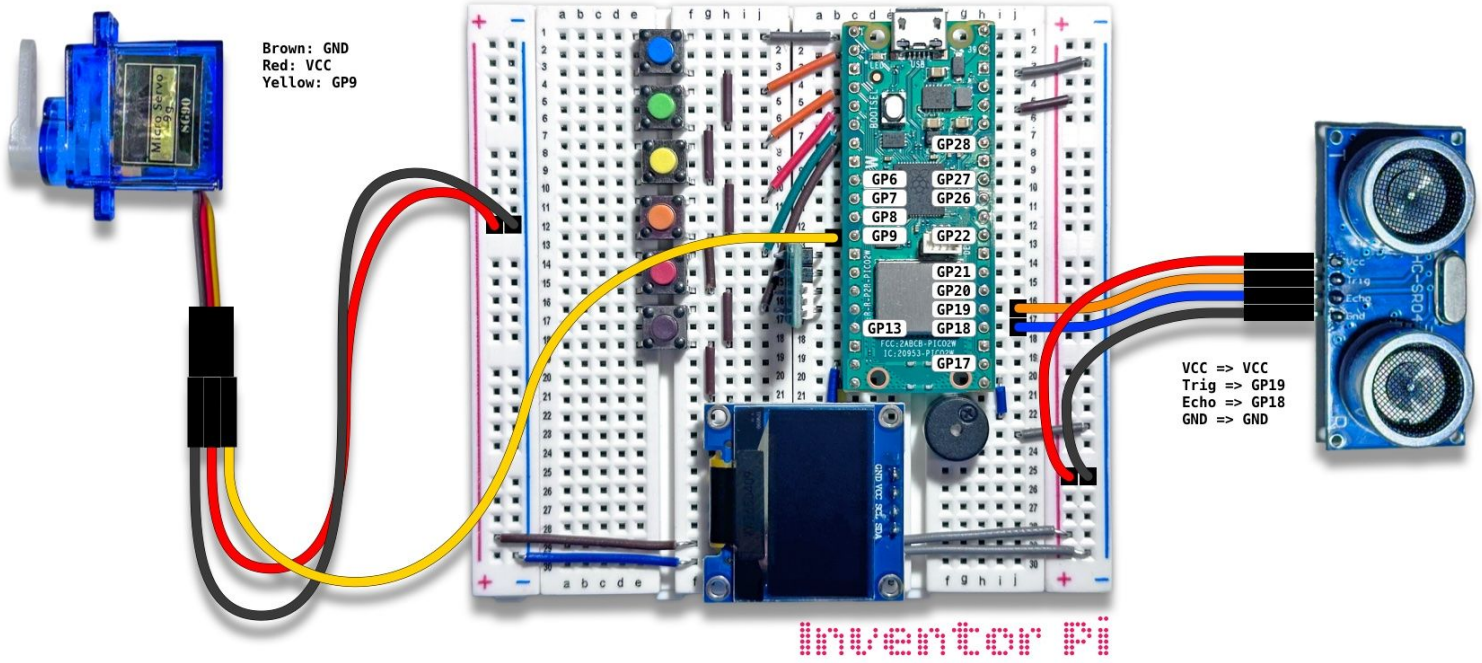
Melody


GP28
GP6
GP27
GP7
GP26
GP8
GP9
GP22
GP21
GP20
GP19
GP13
GP18
GP17
GND VCC SCL SDA
Inventor Pi

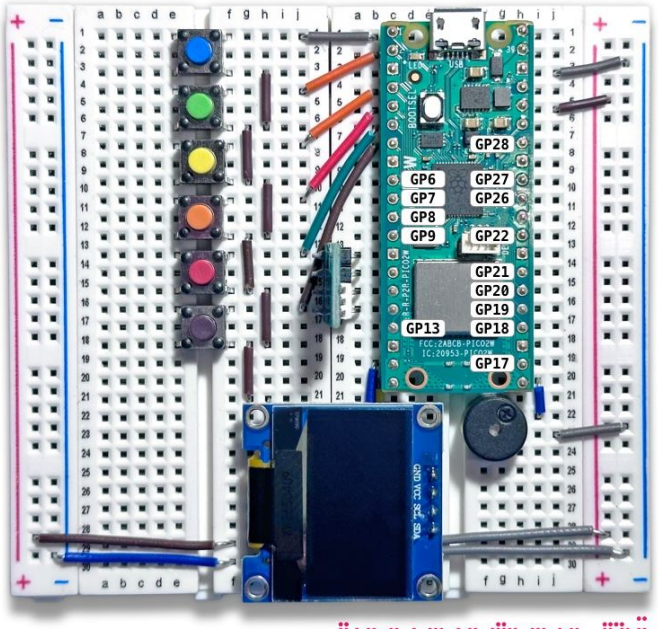
Touch Game


GP28
GP6
GP27
GP7
GP26
GP8
GP9
GP22
GP21
GP20
GP19
GP13
GP18
GP17
GND => GND
VCC => VCC
SIG => GP28
Inventor Pi

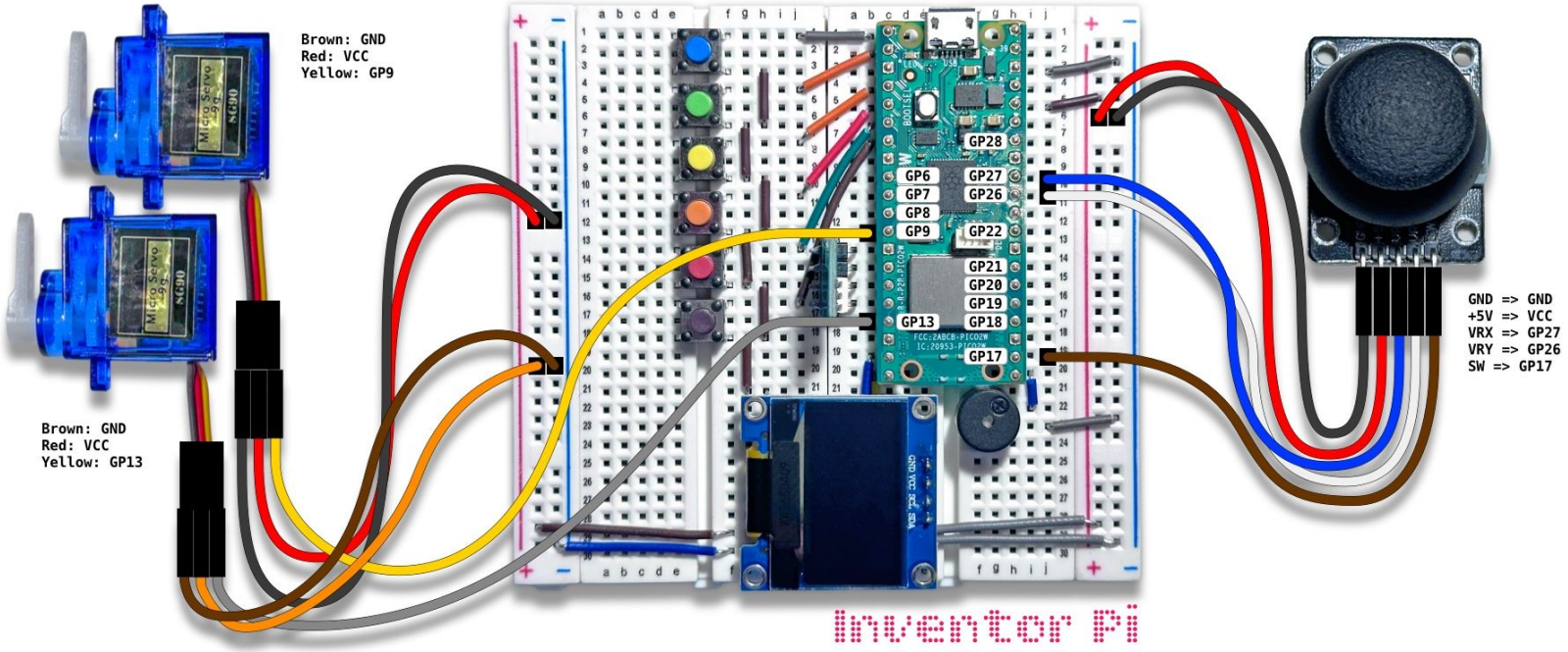
Distance Sensor



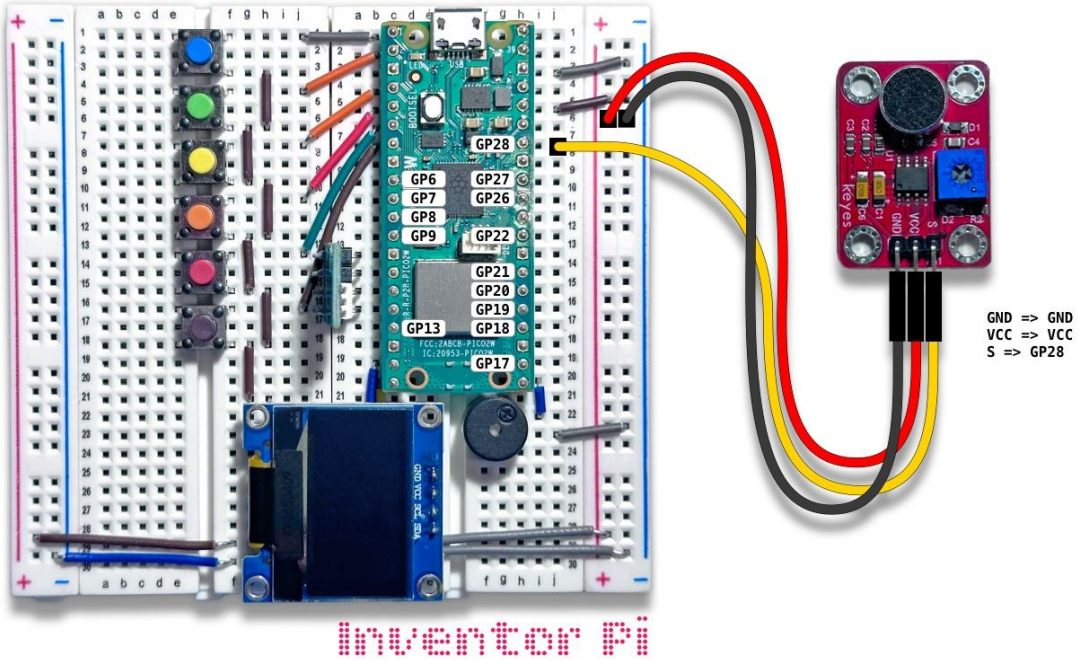
Piano


GP28
GP6
GP27
GP7
GP26
GP8
GP9
GP22
GP21
GP20
GP19
GP13
GP18
GP17
GND VCC SCL SDA
Inventor Pi

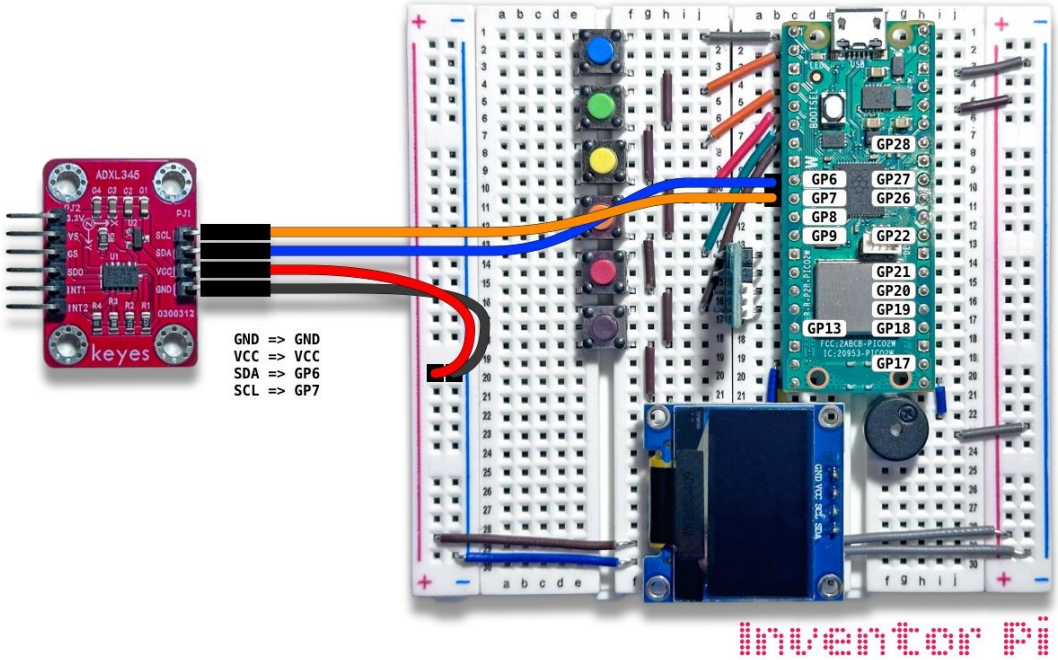
Joystick and Servos



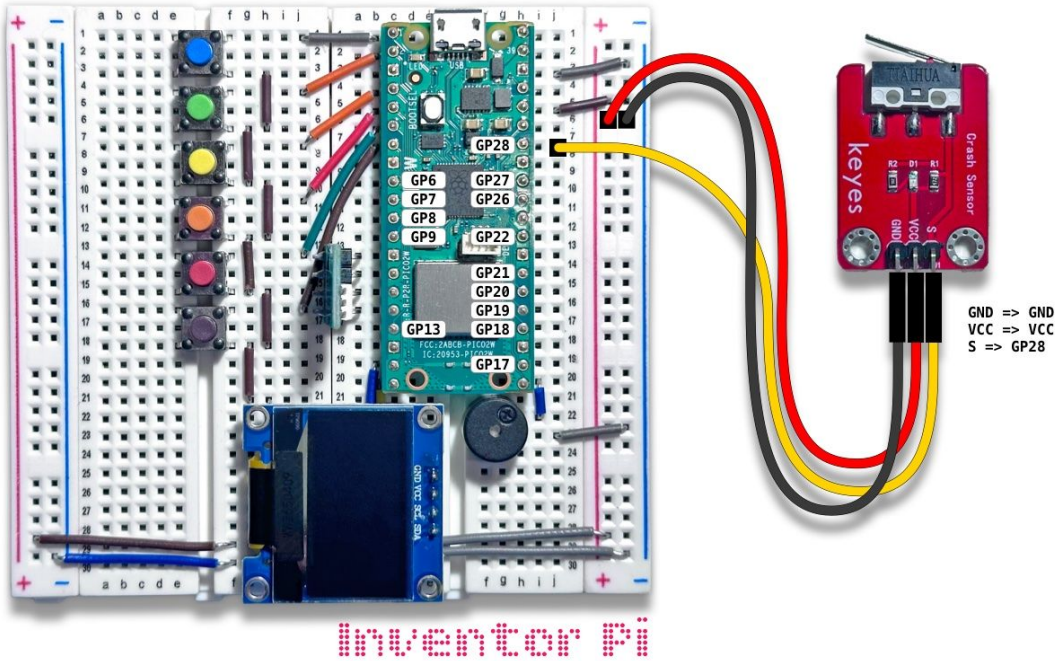
Microphone Sensor



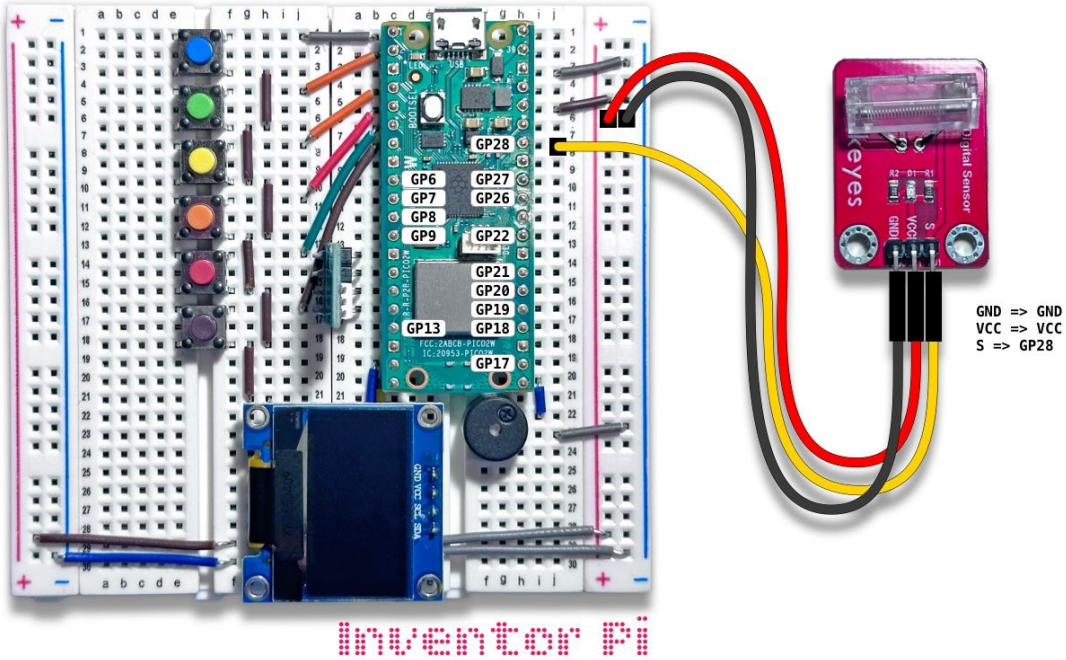
Accelerometer



Crash Sensor



Knock Sensor



Motion Sensor

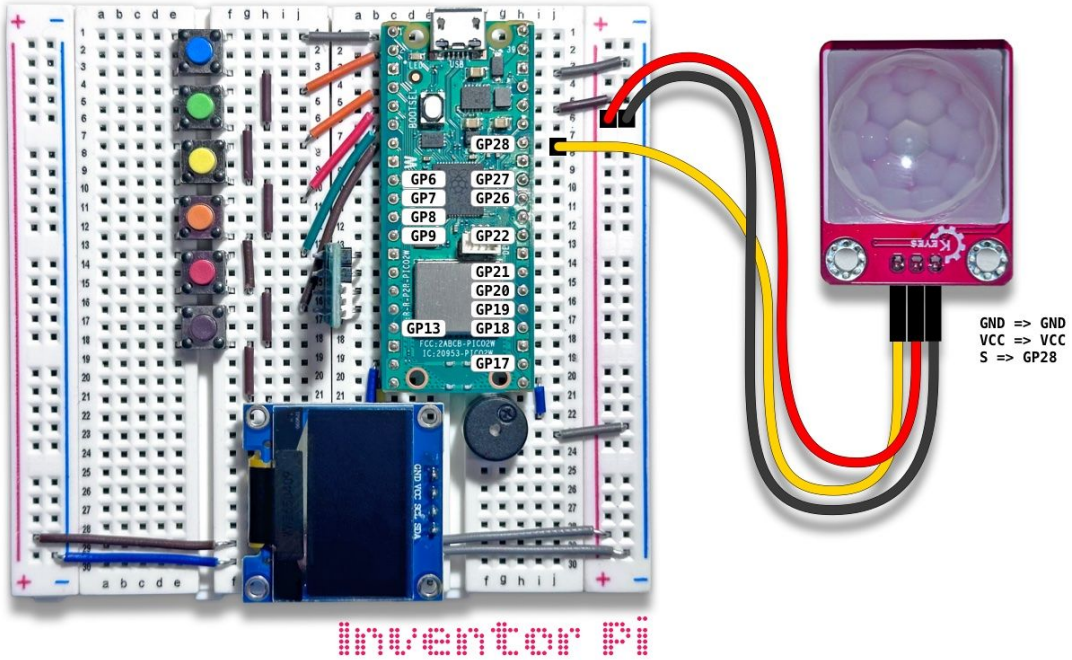
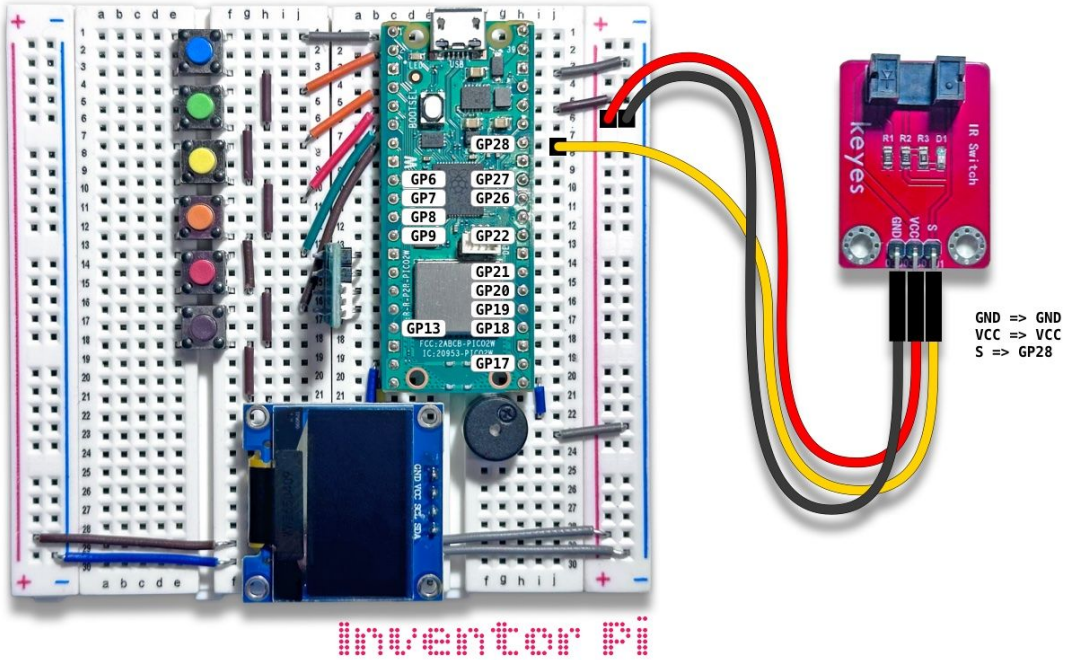
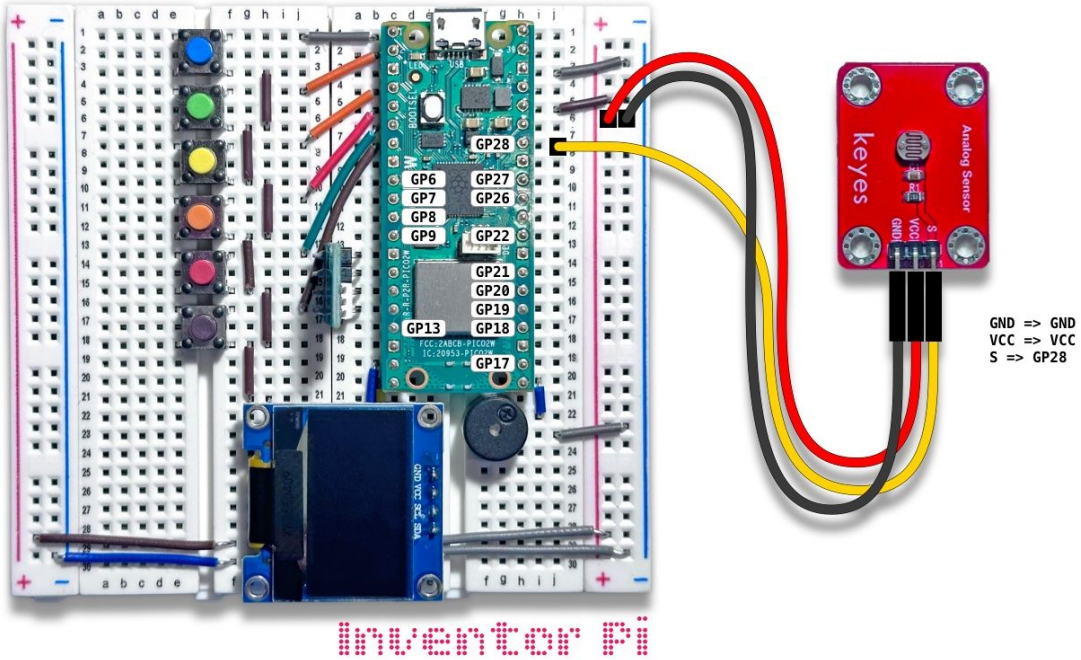


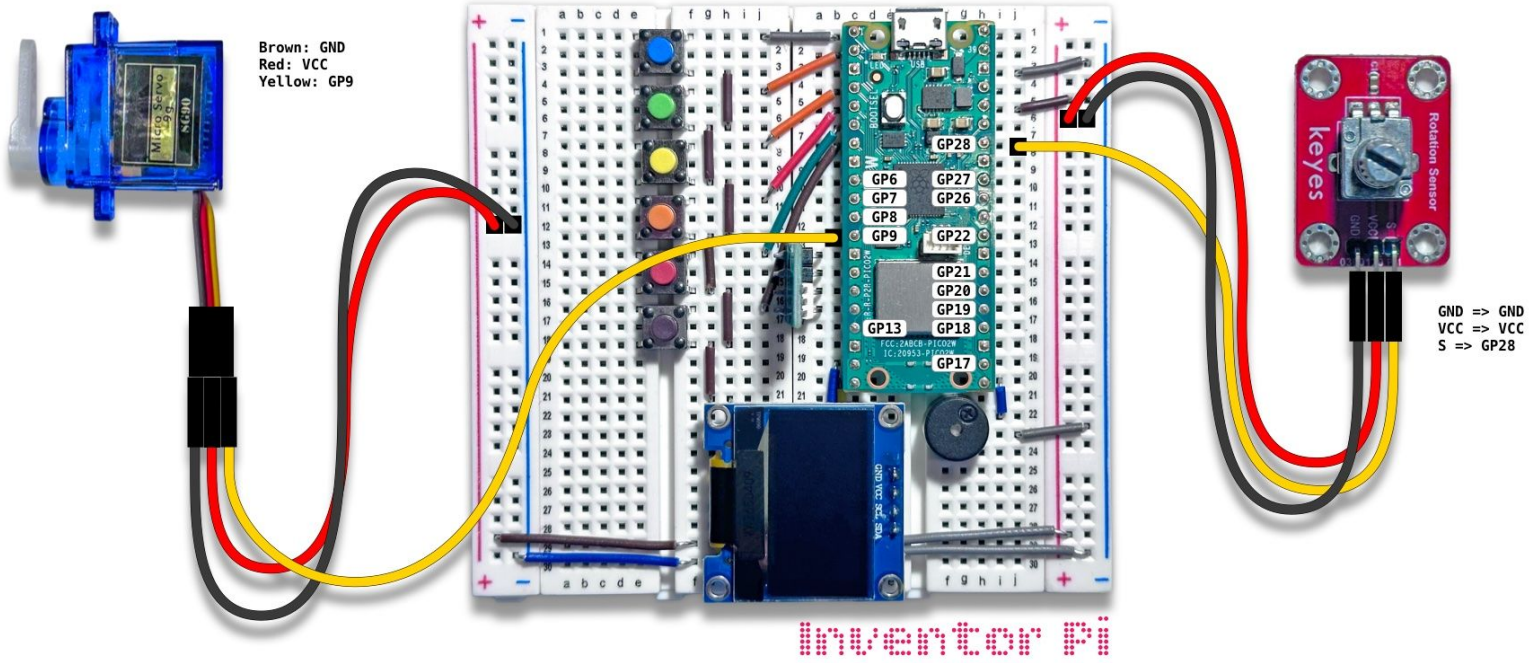
Photo Interrupter



Photoresistor



Potentiometer and Servo



Bluetooth Advertise

